

Library Lending System - REST API Design Document (Week 3)

API Architecture & Conventions .1

.Architectural Style: Representational State Transfer (REST) -

.Format: JSON over HTTP -

Versioning Strategy: URL-Based Versioning (/api/v1/[resource]) to protect -

.existing clients from breaking changes during schema evolutions

Environment Base URL: {{baseUrl}} configured globally in Postman to point -
:to

Local Environment: http://localhost:5108 -

Endpoints Routing Directory .2

:Here is the verified routing map for the Members primary resource

HTTP Method	Endpoint Path	Success Code	Error Code	Description / Intent
-----:	:------:	:------:	-----:	-----:
GET**	`{{baseUrl}}/api/v1/Members`	`200 OK`	`500 Server **`	Retrieves a list of all active library members
GET**	`{{baseUrl}}/api/v1/Members/{id}`	`200 OK`	`404 Not **`	Retrieves a specific member's profile by ID
POST**	`{{baseUrl}}/api/v1/Members`	`201 Created`	`400 Bad **`	Creates/registers a new member in the database
PUT**	`{{baseUrl}}/api/v1/Members/{id}`	`200 OK`	`404` / **`	Updates an existing member's profile details

DELETE** | `{{baseUrl}}/api/v1/Members/{id}` | `204 No Content` | `404 Not \*\*` |
|.Found` | Deletes a member from the database permanently

Detailed Endpoint Specifications (Verified via Postman) .3 

Get All Members .1 

Request: GET {{baseUrl}}/api/v1/Members -

:Response (200 OK) -

```
]
}
,id": 3"
,"fullName": "saifkatout"
,"email": "saifkatout@gmail.com"
,[], : "memberPhones"
[] : "lendingRecords"
,{
}
,id": 4"
,"fullName": "mohammad"
,"email": "mohammad@gmail.com"
,[], : "memberPhones"
[] : "lendingRecords"
{
[
```

Get Member By ID .2 

Request: GET {{baseUrl}}/api/v1/Members/5 -

Response (404 Not Found - Sad Path): (Triggers gracefully when the ID does -

.(not exist in the SQL database, preventing NullReferenceException

}

, "type": "https://tools.ietf.org/html/rfc9110#section-15.5.5"

, "title": "Not Found"

, "status": 404"

traceId": "00-4fdc82b25a8c0aa7b8bdc2099ba8d28f-aed095fb90545267-"

"00

{

Update Member (PUT) .3 

Request: PUT {{baseUrl}}/api/v1/Members/3 -

:Body -

}

, "fullName": "saifkatout"

"email": "saifkatout@gmail.com"

{

.Response (200 OK - Happy Path): Returns the updated payload -

:Failure Path 1 (404 Not Found) 

Request: PUT {{baseUrl}}/api/v1/Members/5 -

.Response (404 Not Found): Handled when trying to update a non-existent ID -

:Failure Path 2 (400 Bad Request - Validation Error) 

Request: PUT {{baseUrl}}/api/v1/Members/3 -

:Body with invalid type (Numeric instead of String) -

}

,fullName": 1221"

"email": "user@example.com"

{

Response (400 Bad Request): Handled by automatic model validation before -

.touching the database

}

, "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1"

, ".title": "One or more validation errors occurred"

, "status": 400"

} : "errors"

] : "fullName.\$"

"...The JSON value could not be converted to System.String"

[

{

{

Delete Member (DELETE) .4 

Request: DELETE {{baseUrl}}/api/v1/Members/6 -

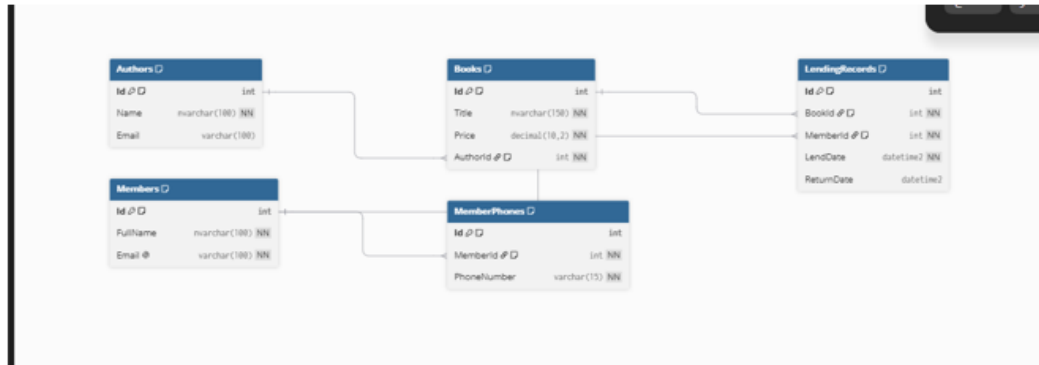
Response (204 No Content - Happy Path): (The record is deleted from SQL -

.(Server successfully. No body is returned, saving network bandwidth

:Failure Path (404 Not Found) 

Request: DELETE {{baseUrl}}/api/v1/Members/44 -

Response (404 Not Found): Handled when attempting to delete an ID that has -
.already been removed or does not exist



Search PUT Upd. DEL Dele GET Memb GET New R GET New R GET New R No environment

week03 > Day04 > Member Save Share

GET http://localhost:5108/api/v1/Members Send

Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body 200 OK 1.80 s 347 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "id": 3,
4     "fullName": "saifikatout",
5     "email": "mohammad@gmail.com",
6     "memberPhones": [],
7     "lendingRecords": []
8   },
9   {
10    "id": 4,
11    "fullName": "mohammad",
12    "email": "mohammad@gmail.com",
13    "memberPhones": [],
14    "lendingRecords": []
15  }
]
```

New Chat

Stabilize Member URL + retest

Add response checks for Member

Fix Day04 requests' base URL

Describe what you need. Press (for context, / for Skills.

Auto

Globals Vault Tools

Search PUT Upd. DEL Delete GET Member GET Member GET NEW R GET NEW R No environment

week03 > Day04 > Member By id Save Share New Chat

GET http://localhost:5108/api/v1/Members/5 Send

Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body 404 Not Found 134 ms 325 B Save Response

JSON Preview Debug with AI

```
1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.5",
3   "title": "Not Found",
4   "status": 404,
5   "traceId": "00-4fdc82b25a8c0aa7b8bdc2099ba8d28f-aed095fb90545267-00"
6 }
```

Save and re-test Mer Add tests for missing Create a data-driven Describe what you

week03 > Day04 > update member Save Share

PUT http://localhost:5108/api/v1/Members/5 Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "fullName": "wasef",
3   "email": "user@example.com"
4 }
```

Body 404 Not Found 101 ms 325 B Save Response

JSON Preview Debug with AI

```
1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.5",
3   "title": "Not Found",
4   "status": 404,
5   "traceId": "00-9e21f83f91846ad3ba7aff0c652a83aa-eb75f6396ae056e7-00"
6 }
```

Get. • GET Sear • PUT Upd. • DEL Dele • GET Memb • GET Men • PUT update + No environment

week03 > Day04 > update member Save Share

PUT http://localhost:5108/api/v1/Members/3 Send

Docs Params Authorization Headers 8 Body • Scripts Tests Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "fullName": 1221,
3   "email": "user@example.com"
4 }
```

Body 400 Bad Request • 34 ms • 545 B Save Response

{ } JSON Preview Debug with AI

```
1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3   "title": "One or more validation errors occurred.",
4   "status": 400,
5   "errors": {
6     "request": [
7       "The request field is required."
8     ],
9     "$.fullName": [
10      "The JSON value could not be converted to System.String. Path: $.
11      fullName | LineNumber: 1 | BytePositionInLine: 18."
12    ]
13  },
14  "traceId": "00-1275789e8e78dd9c7a7c1b1d7976ec9e-16baf7974ab17724-00"
```

Get. • GET Sear • PUT Upd. • DEL Dele • GET Memb • GET Men • PUT updi: + No environment

week03 > Day04 > update member Save Share

PUT http://localhost:5108/api/v1/Members/3 Send

Docs Params Authorization Headers 8 Body • Scripts Tests Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "fullName": "saifkatout",
3   "email": "saifkatout@gmail.com"
4 }
5
6
```

Body 200 OK • 248 ms • 249 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": 3,
3   "fullName": "saifkatout",
4   "email": "saifkatout@gmail.com",
5   "memberPhones": [],
6   "lendingRecords": []
7 }
```

Search PUT Upd. DEL Dele GET Memb GET Men PUT updi DEL Dele + No environment

week03 > Day04 > Delete Member Save Share

DELETE http://localhost:5108/api/v1/Members/6 Send

Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body 204 No Content 60 ms 81 B Save Response

Raw Preview Visualize

```
1
```

Search PUT Upd. DEL Dele GET Memb GET Men PUT updi DEL Dele + No environment

week03 > Day04 > Delete Member Save Share

DELETE http://localhost:5108/api/v1/Members/44 Send

Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body 404 Not Found 18 ms 325 B Save Response

{ } JSON Preview Debug with AI

```
1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.5",
3   "title": "Not Found",
4   "status": 404,
5   "traceId": "00-2385b81f520a745108f3bb4ea4d727fa-810ec0f3ce3047ac-00"
6 }
```


HTTP week03 > Day04 > New Request Saving... Share

DELETE http://localhost:5108/api/v1/Members/7 Send

Docs **Params** Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body 204 No Content • 61 ms • 81 B • Save Response ...

Raw Preview Visualize

1

men • PUT updi • DEL Delete • DEL Delete • GET GET • New En • **GET Get** • New Environment

HTTP week03 > Day04 > Get Use variable Save Share

GET {{baseUrl}}/api/v1/Members Send

Docs **Params** Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body 200 OK • 41 ms • 349 B • Save Response ...

{} **JSON** Preview Visualize

```
1 [
2   {
3     "id": 3,
4     "fullName": "saifkatout",
5     "email": "saifkatout@gmail.com",
6     "memberPhones": [],
7     "lendingRecords": []
8   },
9   {
10    "id": 4,
11    "fullName": "mohammad",
12    "email": "mohammad@gmail.com",
13    "memberPhones": [],
14    "lendingRecords": []
```

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